

1. Administrative & Study Identification

Full Study Title: A Study of Patritumab Deruxtecan in Pediatric Participants With Relapsed or Refractory Solid Tumors (MK-9999-01C/LIGHTBEAM-U01) **Short Title/Acronym:** LIGHTBEAM-U01 Substudy 01C **Protocol Number:** MK-9999-01C (9999-01C) **NCT Number:** NCT06941272 **Posting ID:** 495294456705296974 **Trial Status:** RECRUITING **Sponsor/Company:** Merck (Merck Sharp & Dohme LLC) **Investigational Product:** Patritumab Deruxtecan (HER3-DXd / MK-1022) **Phase of Development:** Phase 1 / Phase 2 **Medical Monitor:** Merck Sharp & Dohme LLC Clinical Director

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** * **Part 1 (Safety Lead-in):** To demonstrate a tolerable safety profile and confirm a preliminary recommended phase 2 dose (RP2D). * **Part 2 (Efficacy Evaluation):** To evaluate the preliminary efficacy of Patritumab Deruxtecan by measuring Objective Response Rate (percentage of participants achieving Complete Response [CR] or Partial Response [PR]).

Secondary Objectives: To characterize the pharmacokinetics (PK) of HER3-DXd over time and further evaluate additional efficacy and safety outcome measures, including Disease Control Rate.

2.2 Background & Rationale To address the significant unmet medical need in pediatric populations with hepatoblastoma (a common liver cancer in young children) or rhabdomyosarcoma (RMS; a cancer starting in muscle cells) whose disease has relapsed or is refractory to standard therapies. Patritumab deruxtecan (HER3-DXd) is a novel antibody-drug conjugate (ADC) designed to attach to specific proteins on cancer cells to deliver targeted cytotoxic therapy. This study evaluates if children can safely tolerate the drug, how their bodies process it, and if it can successfully shrink or eliminate the cancer.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Single-arm (Dose escalation/evaluation followed by expansion) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration **Target N\$:** 50 participants **Number of Sites:** Approximately 25 global locations (including the

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 1 month to 17 years at the time of enrollment. 2. Has one of the following histologically confirmed advanced or metastatic solid tumors: Rhabdomyosarcoma (RMS) or Hepatoblastoma. 3. Has progressed after at least 1 prior systemic treatment for RMS or hepatoblastoma and who has no satisfactory alternative treatment option (i.e., is ineligible for other standard treatment regimens). 4. Participants who have adverse events (AEs) due to previous anticancer therapies must have recovered to Grade ≤ 1 or baseline. (Note: Exceptions exist for adequately treated endocrine-related AEs, Grade ≤ 2 neuropathy, and Grade ≤ 2 alopecia). 5. Hepatitis B surface antigen (HBsAg) positive participants are eligible if they have received hepatitis B virus (HBV) antiviral therapy and have an undetectable HBV viral load. 6. Participants with a history of hepatitis C virus (HCV) infection are eligible if HCV viral load is undetectable.

4.2 Exclusion Criteria 1. Has a history of (noninfectious) interstitial lung disease (ILD)/pneumonitis that required steroids or has current ILD/pneumonitis, and/or suspected ILD/pneumonitis that cannot be ruled out by standard diagnostic assessments. 2. Has clinically severe respiratory compromise resulting from intercurrent pulmonary illness. 3. Has a history of solid organ transplant or allogeneic stem cell transplant. 4. Has clinically significant corneal disease. 5. Known active central nervous system (CNS) metastases and/or carcinomatous meningitis/leptomeningeal disease (participants with previously treated brain metastases may participate provided they are radiologically stable without evidence of progression for at least 4 weeks). 6. Has uncontrolled or significant cardiovascular disorder, or a history of clinically significant congenital cardiac syndrome. 7. Has a history of human immunodeficiency virus (HIV) infection. 8. Has a known additional malignancy that is progressing or has required active treatment within the past 1 year. 9. Has an active infection requiring systemic therapy. 10. Has concurrent active hepatitis B (HBsAg positive and/or detectable HBV DNA) and HCV (anti-HCV antibody positive and detectable HCV RNA) infection. 11. Has not adequately recovered from major surgery or have ongoing surgical complications.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Patritumab Deruxtecan (HER3-DXd) administered once every 3 weeks (21-day cycles). The specific dose will be established during the Part 1 dose-escalation phase to confirm the RP2D. **Route of**

Administration: Intravenous (IV) infusion. **Duration of Treatment:** Until disease progression, unacceptable toxicity, or withdrawal from the trial.

5.2 Concomitant & Prohibited Therapy Permitted: Routine supportive care, hormone replacement therapy for existing endocrine AEs, and stable antiviral therapies for participants with controlled Hepatitis B or C. **Prohibited:** Other concurrent investigational agents or systemic anti-cancer therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Histologic verification, Baseline Imaging, Safety Labs
Baseline	Cycle 1, Day 1	First Dose of HER3-DXd, Baseline PK sampling	Interim Every 21 Days Treatment administration, Safety Labs, Efficacy Assessment, PK monitoring, AE/DLT tracking
End of Study	Up to 5 years	Treatment Discontinuation Visit, Safety Follow-up, Long-term Survival Follow-up	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * **Part 1:** Percentage of participants who experience Dose-limiting Toxicities (DLTs) during Cycle 1 (up to 21 days). * **Part 1:** Percentage of participants who experience an Adverse Event (AE). * **Part 1:** Percentage of participants who discontinue study treatment due to an AE. * **Part 2:** Objective Response Rate (ORR) assessed by the percentage of participants achieving Complete Response (CR) or Partial Response (PR).

7.2 Secondary & Exploratory Endpoints * **Part 2:** Percentage of participants who experience an AE. * **Part 2:** Percentage of participants who discontinue study treatment due to an AE. * **Part 1 & Part 2:** Disease Control Rate (DCR). * Pharmacokinetics (PK) profile of Patritumab Deruxtecan (HER3-DXd) over time. * Duration of Response, Progression-Free Survival (PFS), and Overall Survival (OS).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring for AEs and DLTs (with a focus on identifying interstitial lung disease/pneumonitis early). **Serious Adverse Events (SAEs):** Evaluated and reported per standard 24-hour safety reporting timelines. **Data Monitoring Committee (DMC):** A safety review committee will evaluate DLTs in Part 1 to determine dose escalation and confirm the RP2D.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Analysis Set (all enrolled participants receiving \$\\ge 1\$ dose of study drug) and Efficacy Analysis Set (Intent-to-Treat for relevant cohorts). **Sample Size Justification:** A target of \$N=50\$ provides sufficient power to assess DLT safety thresholds in Part 1 and estimate the preliminary efficacy (ORR) bounds in Part 2 expansion cohorts.

Handling of Missing Data: Standard survival analysis methodologies (e.g., Kaplan-Meier limits and data censoring) will be utilized for time-to-event efficacy endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Standard risk-based monitoring (onsite and remote data verification) to ensure compliance with the clinical protocol and pediatric safety guidelines. **Audit Trail:** Utilizing secure electronic Case Report Forms (eCRF) and centralized data clarification forms.

11. Study Identifiers & Governance

Primary ID: 9999-01C / **MK-9999-01C Registry Number:** NCT06941272

Posting ID: 495294456705296974 **Sponsor/Lead Organization:** Merck

(Merck Sharp & Dohme LLC) **Study Title:** LIGHTBEAM-U01 Substudy 01C

Official Regulatory Title: LIGHTBEAM-U01 Substudy 01C: A Phase 1/2 Substudy to Evaluate the Safety and Efficacy of Patritumab Deruxtecan in Pediatric Participants With Relapsed or Refractory Solid Tumors

12. Clinical Framework (Oncology Specific)

Disease Area / Name: Malignant Neoplasms **Preferred UMLS Name:**

Refractory Solid Neoplasm of Childhood **Primary Condition:**

Relapsed or Refractory Solid Tumors (specifically Hepatoblastoma

and Rhabdomyosarcoma [RMS]) **Diagnosis Threshold:** Histologically confirmed advanced or metastatic disease failing at least 1 prior systemic therapy

Medical Methodology: Antibody-Drug Conjugate

(ADC) targeting HER3 expression **Optimization Target:** Dose determination (RP2D), tumor shrinkage (Objective Response Rate), and Disease Control Rate (DCR)

13. Study Procedures & Methodology

Management Pathway: Pediatric Oncology phase 1/2 pathway

incorporating strict safety lead-ins. **Education Protocol (HCP):**

Extensive training on ADC-related toxicities, specifically identifying and managing suspected interstitial lung disease (ILD) and pneumonitis. **Education Protocol (Patient/Guardian):** Informed consent (and pediatric assent where applicable), treatment expectations, safety reporting, and visit requirements. **Quality Audit Mechanism:** Continuous centralized oversight for AE tracking and DLT adjudication.

14. Observation & Follow-Up Schedule

Total Duration: Up to approximately 5 years **Onsite Visit Cadence:** Day 1 of every 21-day cycle **Remote Monitoring Frequency:** Continuous site-level safety tracking **Checklist Review Interval:** Cycle-by-cycle (Every 3 weeks)

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]					
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]					